

Service Bulletin

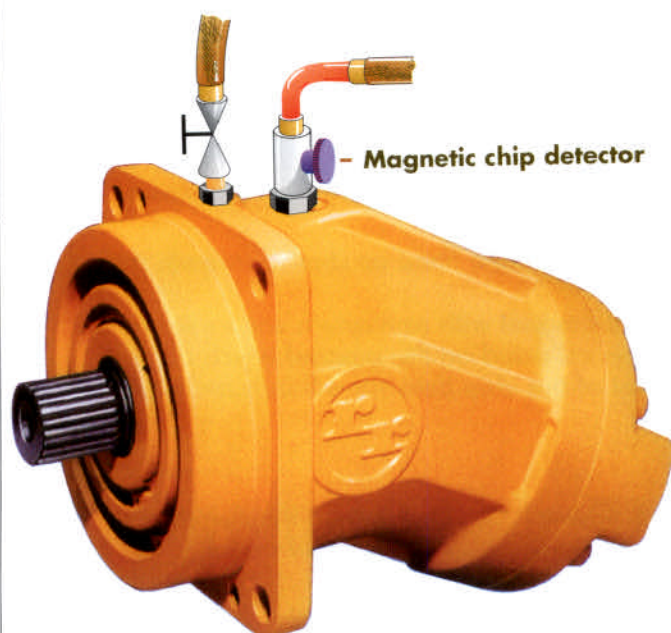
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Improved Bearing Lifetime



Fixed displacement Hydraulic Pumps

Constrained flow through the bearings by FRAMO el. circulation pump unit.



Hydr. powerpack pumps with connections for constrained flow

- ✓ **Considerably increased lifetime.**
- From min. 60% to double.
- ✓ **Cooled oil constantly through bearings.**
- ✓ **Optimal bearing running condition.**
- ✓ **Condition monitoring by magnetic chip detectors.**

GENERAL

Experience has proven that bearing lifetime depends on oil cleanliness and optimum viscosity inside the bearings. To get maximum lifetime on hydraulic pumps a bearing circulation system was introduced as standard from early 1980's.

The new high efficiency hydr. pumps which are replacing the old obsolete ones (ref. Service Bulletin no. 2) require higher oil flow through the bearings to get max. lifetime out of the unit.

For those who want an independent drive we now introduce an electric driven circulation unit. When using electric drive additional features can also be implemented as described below.

DESCRIPTION

A small electric motor driven circulation unit provides the required flushing oil and consists of: Electric motor, hydraulic pump and filter.

Size of power pack and flow rate is dependent of actual vessel installation with regard to total number of hydraulic power packs.

As the Flushing Unit consists of a constant speed el. motor and a fixed displacement pump, constant oil flow is secured during all conditions.

- Balancing equal oil flow on each branch is achieved by one adjustable valve on each pump unit.
- Flushing oil outlet/drain is collected by a common drain line and led upstream of the main filter.
- Start/stop function is automatic, activated by a pressure switch in the main hydraulic pressure line. Set point 20 - 30 bar.
- Filter on the circulation unit ensure clean oil to the bearings.

CONDITION MONITORING

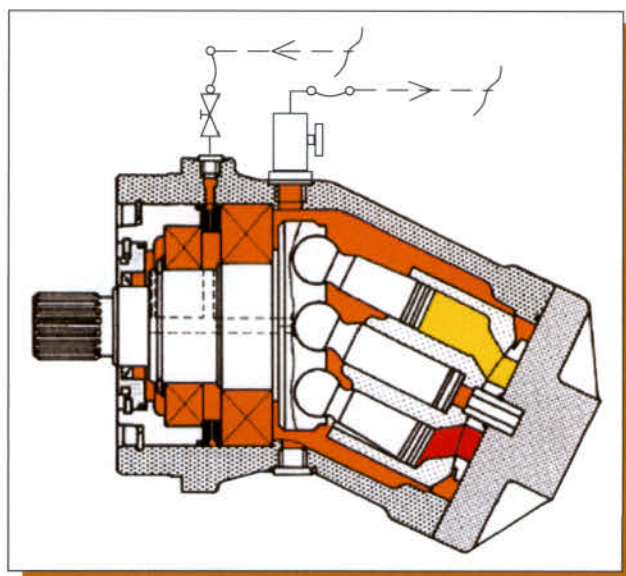
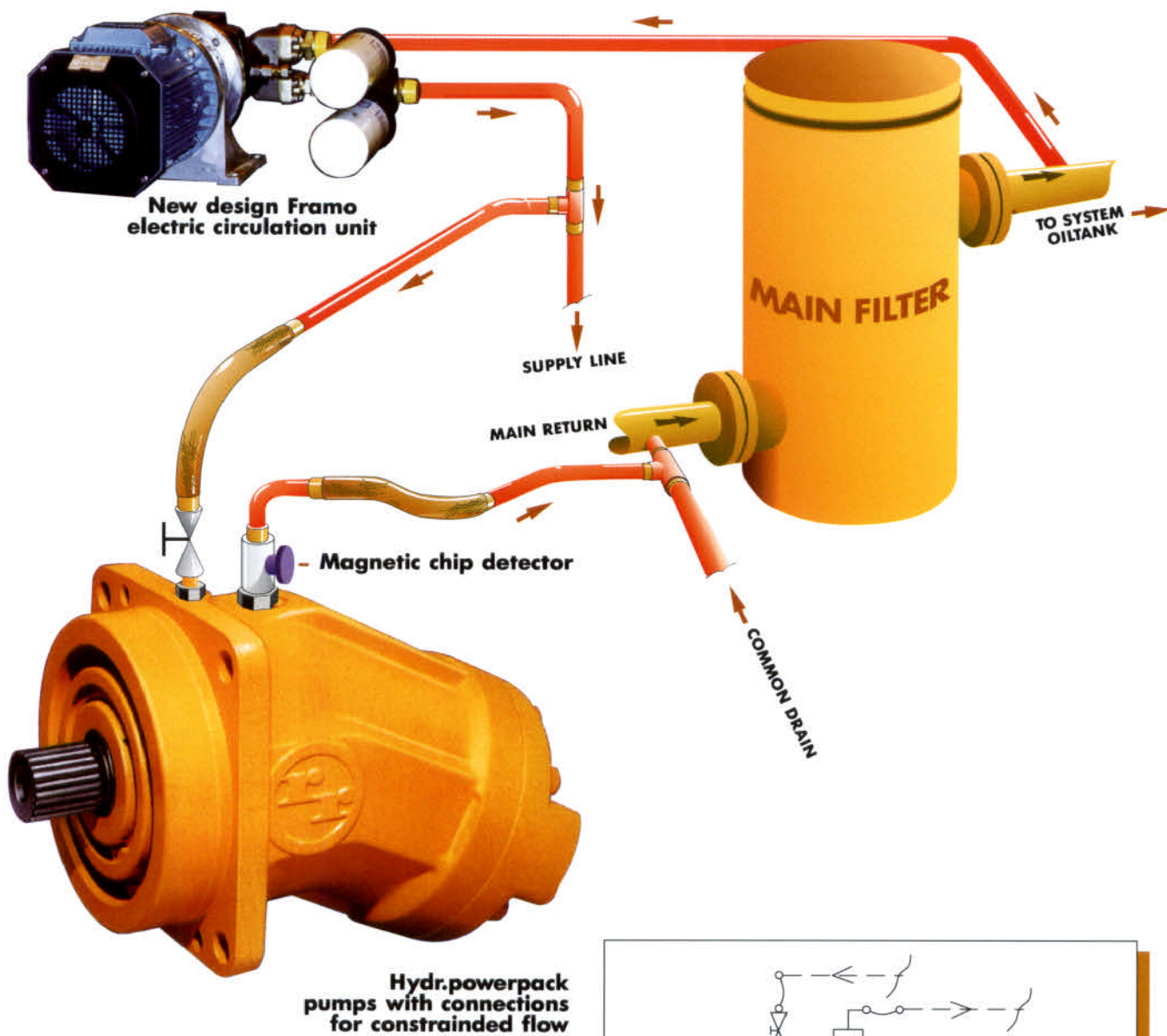
- We recommend that **a magnetic chip detector** should be fitted as shown on each pump.
 - This assembly consist of a snap-on permanent magnet with non return valve into the circulation/drain oil. Any ferritic particle will be collected on the magnet which can be visually inspected at any time. Periodic control of this chip detector will indicate when bearing life comes to an end.
- **OPTION**
Additionally to above one electro magnetic unit w/switch can be fitted on the common flushing/drain pipe, giving alarm signal for "excessive wear" on the control panel.

ADDITIONAL APPLICATION FOR THE ELECTRIC DRIVEN CIRCULATION UNIT

Other benefits for ease of the maintenance can be achieved by hooking-up the circulation unit to the hydraulic storage tank for:

- Correct oil filling and reduce the time for air venting.
- Reduce the work when draining/filling of the main system.
- Reduce time for draining/filling during change of filter cartridges.
- During voyage, filtering the oil in the system tank via the built-on, and main filters.

PRINCIPAL ARRANGEMENT



Yours Faithfully
pr. Frank Mohn Services AS

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